

Structural Replay in Dense Transformers

Under a Frozen FP32 Regime:

Evidence from Closed 3B/4B Models and Boundary Results

Aoi Kawasaki

April 9, 2026

Abstract

Under a frozen FP32 dense-transformer regime, we evaluate replay structure on teacher-forced Gate8 candidate-execution bundles using a fixed Gate12A transport-and-loop audit surface. The main evidence consists of 9 closed runs (3 models $\times$ 3 rendering families) under transformers 4.57.3. The closed lines are Qwen/Qwen2.5-3B-Instruct, meta-llama/Llama-3.2-3B-Instruct, and Qwen/Qwen3-4B. All 9 runs receive a quartet-positive structural replay verdict under the frozen rule.

By contrast, the present first-pass phenotype layer remains family-conditioned and model-specific. The clearest localized concentration within the current closed 3B/4B evidence set appears only in the archive high-tension band (6 packets per model). We also report 4 supporting transcript-only dense-transformer extensions, 2 frozen-protocol admission failures, and 1 non-transformer sidecar (`state-spaces/mamba-2.8b-hf` under an altered observation surface) that does not preserve the current dense-transformer transcript-path pattern. These results should be read as a controlled replay-analysis report under one fixed protocol, not as a free-generation comparison, a benchmark-ranking paper, a mechanistic-causality account, or a universal invariance claim.

1 Introduction

Recent mechanistic-interpretability work has made substantial progress in identifying circuit-level structure and in partially automating circuit discovery (e.g., [1, 2]). More recent work has also begun to localize behavior-specific phenomena such as inflated verbalized confidence [4]. Those lines of work are valuable, but they are typically tied to model-specific internal analyses.

This study does not try to replace that literature with a general alternative. It is also not a free-generation comparison. The reported evidence is built on teacher-forced Gate8 candidate-execution bundles replayed under one fixed Gate12A audit surface, so the object of study is controlled replay structure under a declared protocol rather than open-ended decoding behavior.

Nor is this a benchmark-ranking paper. The custom benchmark is used to keep world truth, retrieval rendering, answer target, and defect labeling jointly explicit under one replay protocol, not to rank general task ability across models.

Within that narrow scope, the mainline result is that the quartet-positive structural replay recurs across the closed 3B/4B dense-transformer full-family-set lines, while phenotype remains family-conditioned and model-specific, with the clearest localized concentration appearing only at archive high tension. Throughout the manuscript, we keep closed full-family-set mainline runs, transcript-level extensions, frozen-protocol admission failures, and altered-surface sidecars analytically separate, and we treat phenotype only as a descriptive overlay on top of an already frozen machine-side structural score.

To study that result, we use an operational auditing surface that treats each run as a replay graph over extracted artifacts. Fixed discrete transport checks and loop-consistency residuals are applied on that graph under one stable protocol surface. The framework does not seek to locate the “cause” of hallucinations within specific weights; rather, it measures whether tensions or broken transport paths recur or fail to recur under the same declared replay conditions.

Additional supporting transcript-level evidence from smaller and larger dense-transformer lines, including one imported Mistral-7B transcript bundle, is reported separately as structural support rather than as mainline phenotype evidence. Boundary checks beyond the mainline show that some candidates fail admission at the protocol boundary, while a non-transformer sidecar (Mamba 2.8B under an altered observation surface) does not preserve the current dense-transformer transcript-path pattern.

1.1 What We Claim

1. **Operational Surface.** We operationalize a fixed transport-and-loop auditing surface for comparing replay artifacts under one frozen protocol.
2. **Closed 3B/4B Structural Replay.** We empirically show quartet-positive structural replay across three closed dense-transformer full-family-set lines under `transformers 4.57.3` and `FP32`.
3. **Bounded Extensions.** We separately report supporting transcript-level evidence on additional dense-transformer lines, including one imported 7B transcript bundle, together with admission-boundary and sidecar results; these extend structural range or delimit the regime, but they are not folded into the mainline phenotype claim.

1.2 What We Do NOT Claim

1. **Universality.** We do not claim this framework is universally applicable to all language models. The Mamba sidecar result under an altered observation surface shows that the current evidence does not support unchanged extension beyond the dense-transformer regime.
2. **Causality of Hallucinations.** We measure the structural manifestation of reasoning conflicts (residuals), but we do not claim to explain the mechanistic causality of why specific

attention heads generate those conflicts.

3. **Scale Independence.** All empirical evidence is strictly bounded to the frozen FP32 execution setting considered here. We make no claims regarding the preservation of these structural invariants in massive-scale models (>70B) or MoE architectures without further validation.

2 Controlled Experimental Regime

The empirical claim advanced here depends on a deliberately narrow controlled regime. The goal is not to maximize model scale, throughput, or benchmark coverage, but to hold the observation surface fixed tightly enough that cross-model comparisons remain interpretable.

2.1 Execution Regime

All mainline evidence in this study is restricted to a frozen FP32 setting. In the archived closed 3B/4B mainline candidate-execution manifests, eight Gate8 runs record CPU execution and one records CUDA execution on L4. In particular:

- candidate generation is executed without quantization
- the observation surface is held fixed at the current attention-weighted extractor
- replay is performed from replayable Gate8 handoff bundles rather than from partial memo-only imports
- no architecture-specific probing, manual carve-out, or post-hoc evaluator adjustment is introduced per model

Operationally, the pipeline is split into two stages:

1. candidate execution at Gate8
2. local deterministic replay and packet construction at Gate12A

This staging split is operational rather than ontological: candidate execution and Gate12A replay are archived as separate stages, but the scientific object is unchanged. In all cases the relevant unit is the same replayable `candidate_execution` artifact, and the present comparison treats that artifact as the unit of evidence regardless of whether the archived candidate-execution manifest records CPU or CUDA execution.

Because the replayable `candidate_execution` artifact is the scientific unit, Gate8 execution controls are treated as part of the frozen protocol rather than as free per-model tuning knobs. In the archived mainline path recovered here, candidate execution is a teacher-forced extraction

path rather than a stochastic decoding path: `extraction_mode = teacher_forcing_forward_v1`, `seed = 7`, `topk_requested = 128`, and `deterministic_requested = true`. The observation-surface definitions are likewise fixed at `Splus = attn_lastlayer_weighted_hidden_v1` and `Sminus = lm_head_row_expectation_topk128_v1`. Those controls were frozen per run for the comparisons reported here and were not retuned per model.

The manuscript therefore makes a deliberate division of labor: the prose states the comparison logic and the exact mainline settings that govern the reported path, while the archived per-run manifests and per-sample meta records remain the source of truth for the full literal Gate8 field record attached to each replayable bundle. Because the reported mainline path is teacher-forced extraction over fixed answer targets, stochastic decoding parameters such as temperature, top- p , and `do_sample` are not applicable to that path. This choice isolates replay structure from decoding stochasticity and lets the report ask whether the same transport-and-loop pattern survives when answer targets are held fixed.

Here, Gate8 refers to the candidate-execution artifact generation stage, and Gate12A refers to the deterministic replay and packet-construction stage that consumes the replayable bundle. Here, a packet denotes one replay-level summarized unit emitted by Gate12A. It is derived from a replayable `candidate_execution` bundle and used for downstream structural summaries and later descriptive overlay. A first-pass phenotype attachment denotes the author-coded descriptive label layer added only after the machine-side structural scoring has already been frozen. A sidecar denotes a run performed under an explicitly altered observation surface.

For reader convenience, Table 1 fixes the small set of protocol terms that recur throughout the rest of the report.

Table 1: Reader-guidance terminology used throughout the report.

Term	Working meaning in this report
Gate8	Candidate-execution stage that produces the replayable <code>candidate_execution</code> bundle.
Gate12A	Deterministic replay stage that consumes the replayable bundle and produces the packet-layer structural surface to which first-pass attachments may later be added.
Packet	Replay-level summarized unit emitted by Gate12A from a replayable <code>candidate_execution</code> bundle and used for downstream structural summaries and later descriptive overlay.
Family	One rendering of the fixed benchmark construction: <code>transcript</code> , <code>briefing</code> , or <code>archive</code> .
Full-family-set closure	A model line for which all three families reach replayable Gate12A closure under the frozen protocol.
Transcript-level extension	A supporting line observed on the <code>transcript</code> family only; useful for structural breadth but not equivalent to full-family-set phenotype coverage.
First-pass phenotype attachment	Author-coded descriptive overlay added only after the machine-side structural scoring has been frozen; never used for structural pass/fail.
Sidecar	A run admitted only under an explicitly altered observation surface and therefore analyzed separately from the mainline.

Figure 1 summarizes the protocol pipeline and the structural replay decision rule used in the present study.

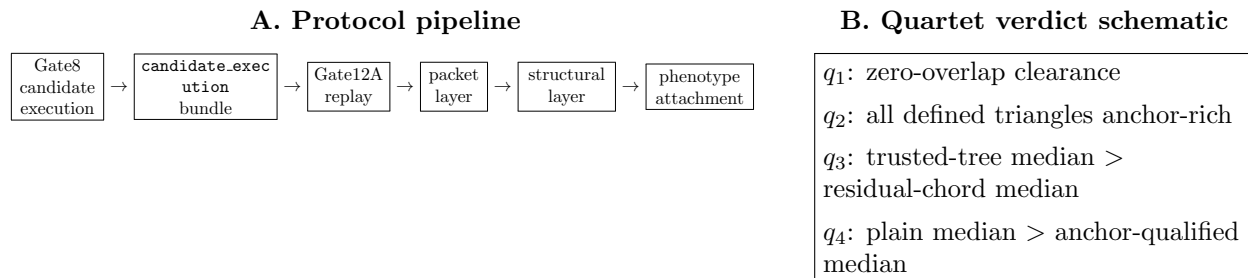


Figure 1: Frozen protocol and structural replay decision rule. Panel A shows the protocol pipeline. Panel B summarizes the machine-side structural quartet. A run is counted as structural replay **only** when all four checks are jointly positive under the frozen protocol.

2.2 Mainline Admission Rule

A model/run is admitted into the dense-transformer main evidence set only if all of the following remain unchanged:

- `transformers==4.57.3`
- FP32 execution
- the existing Gate8 handoff and Gate12A replay protocol
- the current observation surface, i.e. the attention-weighted `Splus` definition
- no extractor branch specific to the candidate model
- no `trust_remote_code` requirement or extra dependency layer beyond the frozen environment

This makes loader success necessary but not sufficient. A candidate may be runnable in a broad engineering sense and still remain outside the main evidence set if it requires a protocol change, a library upgrade, or an observation-surface substitution.

2.3 Mainline Dense-Transformer Surface

Under this admission rule, the present mainline is limited to dense-transformer models that can be run and replayed without altering the frozen protocol. The strongest current evidence comes from closed full-family-set lines on:

- Qwen/Qwen2.5-3B-Instruct
- meta-llama/Llama-3.2-3B-Instruct

- Qwen/Qwen3-4B

with supporting transcript-level structural extension on smaller and larger dense-transformer lines.

Crucially, the manuscript distinguishes between:

- full-family-set closures, which support phenotype comparison
- transcript-level extensions, which broaden the structural range but do not by themselves establish phenotype portability

2.4 Evidence Tiers And Run Counts

The present evidence set comprises nine closed mainline runs (three models $\times$ three families) and four supporting transcript-only runs, one of which is imported. All of these runs are derived from one fixed custom source benchmark line built under `gate8_semiclosed_conflict_batch_v1`; the report does not use an external public benchmark such as MMLU, GSM8K, or HumanEval. At source level, that benchmark line contains 200 rows organized across four explicit cells: `clean_support` (50), `direct_contradiction` (50), `distributed_incompatibility` (50), and `surface_noisy_clean` (50). The closed-world relation types are `genealogy` (68 rows), `temporal` (66 rows), and `reachability` (66 rows). Each source row keeps four layers explicit: world truth, retrieval rendering, answer target, and defect labeling. Appendix A gives a compact construction summary, and Appendix A.1 gives concrete transcript-family examples.

A bespoke benchmark is used here not to optimize for task difficulty or leaderboard comparability, but to keep world truth, retrieval rendering, answer target, and defect labeling simultaneously explicit under one fixed replay protocol. Public benchmark suites were not designed for this exact closed-world transport-and-loop audit setting.

Answer-target regimes are fixed by cell. `clean_support` and `surface_noisy_clean` use `consistent_answer` only. `direct_contradiction` uses `consistent_answer` together with `conflict_following_wrong_answer`. `distributed_incompatibility` uses `consistent_answer` together with `unsupported_bridge_answer`. The three families used in the report—`transcript`, `briefing`, and `archive`—are not different benchmark tasks; they are distinct rendering families over the same 200-row source benchmark.

Table 2 gives a compact evidence map for how the mainline, extension, boundary, and sidecar tiers enter the claim surface of the report.

The current closed 3B/4B mainline artifacts do not realize every family at the same row count. The realized `transcript` artifact has 128 rows, the realized `archive` artifact has 128 rows, and the realized `briefing` artifact has 200 rows. The tables below should therefore be read as replay summaries over these fixed realized artifacts rather than as one undifferentiated 200-row dump. Table 4 reports medians over the fixed packet set for each run/family rendering, while Table 6 reports counts over the fixed six-packet high-tension band, with balanced-per-band fixed at 6 in

Table 2: Evidence-map summary for the claim tiers used in the report.

Tier	Models / cases	Family coverage	Role in claim
Mainline	Qwen/Qwen2.5-3B-Instruct; meta-llama/Llama-3.2-3B-Instruct; Qwen/Qwen3-4B	transcript , briefing , archive	Main structural evidence and bounded phenotype comparison
Transcript-only extension	SmolLM2-360M; meta-llama/Llama-3.2-1B-Instruct; Qwen/Qwen2.5-1.5B-Instruct; imported mistralai/Mistral-7B-Instruct-v0.3	transcript only	Structural range only; not full-family phenotype evidence
Admission failures	google/gemma-4-E2B-it; RWKV/v6-Finch-1B6-HF	not admitted	Delimit the frozen-protocol boundary
Sidecar	state-spaces/mamba-2.8b-hf	altered-surface transcript only	Boundary divergence outside the mainline

the archived manifests. All four extension runs use the **transcript** family only. The imported Mistral-7B line is included only as supporting transcript-level evidence on **transcript_v1**; it is not a locally generated full-family-set mainline run and is not used as full-family phenotype evidence. Structural positive/negative status is assigned only from the quartet pass/fail logic summarized in Section 3.8.

2.5 Exclusions And Sidecars

The regime is also defined by what it excludes.

Some candidates fail admission before any meaningful empirical comparison can be made. For example, a model may require a library version beyond the frozen environment or require remote-code/dependency expansion that the current regime forbids. These are recorded as admission failures, not as negative empirical results.

Other candidates can be run only by explicitly changing the observation surface. These are treated as sidecars rather than as mainline evidence. In the present work, this includes the **Mamba 2.8B** transcript sidecar, which required an attention-free fallback surface (**prefix_mean_hidden_v1**). That run is informative, but it is not part of the dense-transformer mainline and is interpreted separately.

2.6 Why This Regime Matters

This frozen regime is restrictive by design. Its purpose is to ensure that any cross-model agreement is not purchased by silent environment drift, architecture-specific extraction logic, or hidden evaluator flexibility.

The controlled-regime claim is therefore narrow but defensible:

- when structural replay persists inside this regime, the result is stronger because the protocol was not relaxed to obtain it
- when a candidate is excluded, the exclusion is itself informative because it marks a boundary of the current evidence surface

- when a sidecar is admitted under an explicit observation-surface change, that result is interpretable precisely because it is not mixed back into the mainline

The present study should therefore be read as an account of what survives under a frozen FP32 dense-transformer protocol, not as a universal survey of all contemporary language-model architectures.

3 Operational Framework

This section specifies the fixed observation surface and replay criteria used in the current experiments. The goal is not to present a general mathematical theory of reasoning, but to define the operational objects, transport checks, and residual tests on which the present empirical claim depends. The terminology is geometric only insofar as it organizes artifact-level comparisons on a common surface [3].

3.1 Reasoning Trajectories As Finite Artifact Graphs

For each benchmark sample, candidate execution produces a finite sequence of token-indexed extraction rows together with answer-linked anchor material. These artifacts are not treated merely as isolated vector states. Instead, they are assembled into a typed finite graph whose nodes include token states, answer states, and anchor-conditioned support/conflict objects.

The graph is therefore not an abstract latent manifold. It is a concrete artifact graph produced by a fixed extraction and replay pipeline. Its importance is methodological: it lets us ask not only where a model ends, but how adjacent local states remain transport-compatible as the trajectory unfolds.

3.2 Local Objects From Projected Proxy Vectors

At the token level, each extracted row provides a triple of projected proxy vectors:

- V
- S_{plus}
- S_{minus}

Under the frozen mainline, S_{plus} is defined by the attention-weighted last-layer surface. These token-level rows are then aggregated into local objects by a native local-span construction that produces a rank-truncated orthogonal basis from the token-level rows. The resulting local object

for a node is therefore represented by this basis together with its singular spectrum and local coordinates.

For the purposes of the present study, the important point is not the internal linear-algebra implementation itself, but the fact that each node is converted into a comparable finite local object with:

- a basis
- a local rank
- a transport-relevant spectrum

This is the object on which downstream transport is defined.

3.3 Discrete Transport Operators

Given two adjacent local objects, the framework constructs a discrete transport operator by comparing their local bases through overlap and singular-value structure. This defines a basis-aware source-to-target transport relation rather than a scalar similarity alone.

The relevant judgment is not whether two states are “close” in a generic embedding sense. The relevant judgment is whether a transport between their local objects remains admissible under the fixed overlap/SVD law. In this sense, the framework is process-sensitive: it measures whether the trajectory can be stitched coherently from one local object to the next.

3.4 Loop Consistency Residuals

Transport along single edges is not enough. A trajectory can look locally plausible and still fail to close consistently over a loop.

For that reason, the framework searches triangle cycles and computes loop-consistency residuals under the current discrete-connection law, specifically whether the composed transport around a triangle cycle remains within a declared compatibility threshold.

The present report uses this layer operationally rather than theoretically. We claim only that these residuals are well-defined on the current reasoning-trajectory graph and empirically informative under the fixed protocol.

3.5 Trusted-Tree / Residual-Chord Decomposition

Beyond loop residuals alone, the framework decomposes the replay graph into two functionally distinct edge classes:

- trusted-tree candidates
- residual-chord candidates

Intuitively, trusted-tree edges are those retained as the most structurally stable transport backbone under the fixed policy, while residual chords mark bypass structure where residual tension remains visible rather than absorbed into that backbone. This decomposition is not imposed post hoc by hand for each model. It is generated under the same declared policy over the replay artifacts.

This decomposition matters because the current main empirical result is not just that “something geometric happens.” It is that under the frozen dense-transformer regime, the same broad separation between structurally trusted transport and residual tension reappears across multiple model lines.

3.6 Structural Replay Versus Phenotype

The framework distinguishes two empirical layers.

The first is the machine-side structural layer. This is where quantities such as:

- zero-overlap clearance
- all defined triangles anchor-rich
- trusted-tree versus residual-chord separation
- plain-over-anchor-qualified dominance

are measured without human interpretive attachment.

The second is the phenotype layer. This is produced only after packet construction, when a narrow first-pass reading is attached to the high-tension and flat bands.

Those phenotype labels are descriptive summaries assigned only after the machine-side structural layer has already been scored; they are not decision criteria for structural pass/fail.

This distinction is essential. Structural replay can persist even when phenotype does not. In fact, that asymmetry is one of the central empirical findings of the current report.

The structural and phenotype layers described above are interpreted only on a fixed observation surface. Runs obtained under an explicit observation-surface substitution are treated as sidecars rather than as direct extensions of the mainline, and are therefore analyzed separately. The formal admission/exclusion boundary is specified in Section 2, while the boundary outcomes are reported in Section 6.

3.7 Minimal Replay-Scoring Procedure

For reader orientation, the frozen replay-scoring path used in this report is summarized below.

1. extract token-level rows and anchor-linked artifacts from a teacher-forced Gate8 `candidate_execution` bundle;
2. aggregate token-level rows into comparable local objects;
3. evaluate admissible transport between adjacent local objects under the fixed overlap/SVD law;
4. evaluate triangle-cycle compatibility under the fixed holonomy-residual bound;
5. summarize packet-layer quantities and compute the quartet conditions (q_1, q_2, q_3, q_4) ;
6. count a run as structural replay only when all four quartet conditions are jointly positive.

3.8 Protocol Summary

For ease of falsification, the current replay layer is scored under one fixed implementation-level protocol. The numerical summaries reported in the main text are artifact-frozen outputs generated under commit `084eb7878d8cb016243950e1cf4b4bd7379daaba`, as recorded in the archived run manifests and per-sample meta records, with one upstream Gate8 exception: the `Qwen/Qwen2.5-3B-Instruct` transcript bundle was generated under earlier commit `58d06742f23a0bc7ba25c6ecde790e2e03b4324e`. The Zenodo package additionally includes commit `8a14c94b999d823c1734b2582730c3ad4ea98d03` as the archived release-time reference implementation preserving the same replay criteria together with the sidecar-boundary logic. The threshold constants and associated decision rules were fixed before comparative analysis and were not tuned per model or per family in this study. Appendix B records the exact numeric thresholds recovered for the mainline comparison, together with the fixed band-size constant used in the text. Table 3 gives a reader-facing operational gloss for quartet terms that can otherwise read like implementation shorthand. These glosses are interpretive aids for the manuscript rather than replacements for the archived implementation-level definitions.

Table 3: Reader-facing operational gloss for the quartet terms used in the main text.

Term	One-line operational meaning
zero-overlap clearance	No realized packet comparison used by the replay score exhibits nonzero overlap below the frozen admissibility rule.
anchor-rich	Every defined triangle retains at least one anchor-qualified support attachment under the frozen rule.
trusted-tree	Packet-layer summary computed on the transport backbone retained as structurally trusted by the fixed policy.
residual-chord	Packet-layer summary computed on tension-bearing bypass edges not retained in the trusted transport backbone.
plain	Packet-layer summary on the unrestricted packet surface before anchor qualification is imposed.
anchor-qualified	Packet-layer summary restricted to material that satisfies the frozen anchor-qualification rule.

- **transport admissibility:** an edge is counted as transport-admissible only when it meets the implementation’s fixed overlap gate and the associated SVD/rank admissibility check
- **triangle compatibility:** a triangle is counted as compatible only when its composed transport residual remains within the implementation’s fixed compatibility bound
- **structural quartet:** a run is positive on the quartet only when all four machine-side checks are positive together: zero-overlap clearance, all defined triangles anchor-rich, trusted-tree dominance over residual-chord structure, and plain-over-anchor-qualified dominance
- **structural replay:** a run is counted as structural replay only when the quartet is jointly positive under that same fixed protocol

To avoid prose/code drift, this report treats the archived reference implementation together with the per-run manifests and per-sample meta records as the normative record for exact numeric thresholds and literal Gate8 field values. The manuscript therefore states the exact recovered mainline constants and execution settings, while leaving the archived artifacts as the final source of truth for the full field-level record.

3.9 What The Framework Contributes

Under the present study’s narrow scope, the framework contributes three things.

First, it provides a fixed audit surface that complements local mechanistic approaches without attempting to replace them.

Second, it defines a fixed-protocol transport-and-loop-based structural layer that can be compared across multiple dense-transformer lines under one frozen protocol.

Third, it gives a principled way to record exclusions and sidecars without contaminating the main evidence set. This is why the same report can contain:

- quartet-positive structural replay
- phenotype non-portability
- admission failures
- and non-transformer sidecar divergence

without collapsing those into one undifferentiated claim.

4 Main Dense-Transformer Evidence

The main empirical evidence in the present study comes from two layers:

- closed full-family-set lines at 3B/4B scale, which support both structural and phenotype comparison
- transcript-level extensions outside that core set, which broaden the structural range but do not by themselves establish phenotype portability

4.1 Structural Replay Persisted Across The Closed 3B/4B Lines

Under the frozen FP32 regime, three dense-transformer lines were closed at the full-family-set level:

- Qwen/Qwen2.5-3B-Instruct
- meta-llama/Llama-3.2-3B-Instruct
- Qwen/Qwen3-4B

For each of these lines, all three families (**transcript**, **briefing**, **archive**) reached replayable Gate12A closure. Across all nine resulting runs, the machine-side structural quartet was positive:

- zero-overlap clearance
- all defined triangles anchor-rich
- trusted-tree dominance over residual-chord structure
- plain-over-anchor-qualified dominance

Table 4 makes that pattern explicit.

Table 4: Closed 3B/4B full-family-set structural summary. Medians are computed over the fixed packet set for each run/family rendering. The four median columns report trusted-tree, residual-chord, plain, and anchor-qualified medians separately; the replay verdict depends on their fixed ordering relations rather than on an externally calibrated absolute scale.

Model	Family	zero-overlap	all defined triangles anchor-rich	trusted-tree median	residual-chord median	plain median	anchor-qualified median
Qwen/Qwen2.5-3B-Instruct	transcript	True	True	1.062	0.888	1.075	0.853
Qwen/Qwen2.5-3B-Instruct	briefing	True	True	1.079	0.805	1.092	0.822
Qwen/Qwen2.5-3B-Instruct	archive	True	True	1.070	0.857	1.083	0.785
meta-llama/Llama-3.2-3B-Instruct	transcript	True	True	0.985	0.656	0.992	0.600
meta-llama/Llama-3.2-3B-Instruct	briefing	True	True	0.974	0.600	0.989	0.623
meta-llama/Llama-3.2-3B-Instruct	archive	True	True	0.975	0.574	0.986	0.606
Qwen/Qwen3-4B	transcript	True	True	0.976	0.398	0.981	0.464
Qwen/Qwen3-4B	briefing	True	True	1.009	0.487	1.018	0.558
Qwen/Qwen3-4B	archive	True	True	0.988	0.395	0.995	0.448

These packet-layer medians are dimensionless summary values. For the replay verdict, the relevant information is the fixed ordering relation *trusted-tree* > *residual-chord* and *plain* > *anchor-qualified*, not an absolute calibrated scale shared with external benchmarks.

The last four columns are reported separately so that the two dominance checks inside the positive quartet are externally legible. In every positive row shown here, the trusted-tree median exceeds the residual-chord median and the plain median exceeds the anchor-qualified median, making the

ordering behind the reported positive q_3/q_4 outcomes visible directly in the table rather than only in the prose.

This yields nine quartet-positive structural replay verdicts under one frozen protocol. The significance of this result is not that the models agree on phenotype. They do not. The significance is that the same quartet-positive structural replay survives across multiple dense-transformer families and across distinct model lines without protocol relaxation, model-specific probing, or architecture-specific evaluator tuning.

4.2 Transcript-Level Extensions Broaden The Structural Range

Beyond the closed 3B/4B lines, supporting transcript-level evidence was added from additional dense-transformer lines spanning smaller and larger scales, including one imported 7B transcript bundle:

- SmolLM2-360M
- meta-llama/Llama-3.2-1B-Instruct
- Qwen/Qwen2.5-1.5B-Instruct
- mistralai/Mistral-7B-Instruct-v0.3

These supporting transcript-level lines were consistent with the same structural quartet described above. They therefore support a broader transcript-level structural range from 360M upward, including one imported 7B transcript bundle, while remaining inside the dense-transformer family.

Appendix Table 9 summarizes these supporting dense-transformer lines and notes that the machine-side structural quartet remained positive in each case.

However, these extension lines are not used here as phenotype-comparison lines. Their role is narrower. They show that quartet-positive structural replay is not confined to a single vendor or to the 3B/4B range alone. They do not, by themselves, establish phenotype portability across scale.

4.3 Phenotype Did Not Collapse To A Single Cross-Model Template

The phenotype layer reported in this section is exploratory and descriptive. It is attached only after the machine-side structural layer has been scored, and it is included as a reader-facing summary layer rather than as a validated annotation scheme or a criterion for structural pass/fail.

The phenotype layer should therefore be read as a descriptive author-coded overlay on top of an already frozen machine-side structural surface, not as an independently validated annotation protocol. Its role in the present report is reader-facing summarization rather than adjudication of the main structural claim.

All first-pass labels were assigned by the first author in a single non-blind pass after the machine-side structural layer had been finalized. No inter-annotator agreement or adjudication protocol was applied; this labeling is therefore treated as a qualitative descriptive layer rather than a validated annotation.

The phenotype taxonomy used here is a narrow first-pass shorthand rather than a formal ontology:

- **conflict-respected**: the contradiction is carried forward rather than repaired away
- **hypothetical-rejection**: direct commitment is deferred through hypothetical framing
- **phantom-barrier**: a conditional or abstract barrier is introduced instead of direct resolution
- **surface-noise**: the output remains locally plausible but structurally uninformative
- **anchor-overreach**: anchor-derived support is extended beyond its warranted scope
- **compensatory-closure**: unsupported reconciliation is introduced to close tension
- **support-fused**: distinct support strands are merged into one undifferentiated positive stance
- **break-candidate**: the high-tension response collapses toward explicit conflict adoption or structural break

Here and below, the suffix “-only” denotes single-label dominance within the relevant band rather than a distinct base category. In the archive concentration table below, “conflict-adopted break-candidate” denotes a **break-candidate** first-pass reading whose dominant form explicitly adopts the conflict, while **surface-noise-only** denotes single-label **surface-noise** dominance.

Although structural replay persisted, phenotype did not collapse to a universal pattern across the closed 3B/4B lines. The first-pass readings remained family-conditioned and model-specific.

These first-pass labels are qualitative summaries produced after the machine-side structural layer has been evaluated; they are descriptive rather than adjudicative, and they are never used as criteria for structural pass/fail or for establishing the main structural claim of the present study.

Table 5: Closed 3B/4B phenotype matrix (high-tension / flat bands).

Model	Transcript (high / flat)	Briefing (high / flat)	Archive (high / flat)
Qwen/Qwen2.5-3B-Instruct	conflict-respected + hypothetical-rejection / conflict-respected + surface-noise	support-fused + conflict-respected / anchor-overreach + conflict-respected + surface-noise	break-candidate + surface-noise / compensatory-closure + conflict-respected + surface-noise
Llama-3.2-3B-Instruct	phantom-barrier + surface-noise / compensatory-closure + conflict-respected + surface-noise	anchor-overreach + compensatory-closure + support-fused / surface-noise-only	break-candidate + surface-noise / surface-noise-only
Qwen/Qwen3-4B	surface-noise + support-fused / surface-noise-only	surface-noise + conflict-respected / anchor-overreach + conflict-respected + surface-noise	break-candidate-only / surface-noise + support-fused

Several points follow immediately from this matrix.

First, transcript phenotype already diverges sharply across model lines. **Qwen2.5-3B** remains conflict-respected-heavy, **Llama-3.2-3B** shifts toward phantom-barrier structure, and **Qwen3-4B** moves toward surface-noise dominance. This means that even when the structural replay layer remains stable, transcript phenotype is not portable in any simple cross-model sense.

Second, briefing remains similarly non-portable. The three closed lines do not settle into one shared briefing phenotype. Instead they preserve distinct mixtures, including support-fused, anchor-overreach, conflict-respected, compensatory-closure, and pure surface-noise outcomes.

Third, archive is the only family that begins to show a narrower cross-model concentration. Even there, however, the concentration is partial rather than total. The commonality lies primarily in the high-tension band, while the flat band remains model-specific. Because this is the clearest localized concentration point in the present evidence set, it is treated separately in the next section.

4.4 What This Evidence Set Supports

Taken together, the present dense-transformer evidence supports four narrow conclusions.

1. First, structural replay persists across the closed 3B/4B full-family-set lines without relaxing the frozen protocol.
2. Second, transcript-level extensions support the view that this structural persistence is not confined to one scale or one vendor line, and supporting imported transcript evidence reaches as high as 7B.
3. Third, phenotype does not inherit the same portability. Structural replay can persist even when phenotype remains family-conditioned and model-specific.
4. Fourth, the present evidence should be read asymmetrically: broad on the structural side, but intentionally narrower on the phenotype side. The strongest phenotype claims in the present study therefore come from the closed 3B/4B lines, while the broader transcript-level lines serve to extend structural range rather than to imply full phenotype universality.

These conclusions are intentionally asymmetric: strongest at the machine-side structural layer, narrower at the phenotype layer, and explicitly bounded at the protocol boundary.

5 Archive High-Tension Localized Concentration

The closed 3B/4B dense-transformer lines do not share a universal phenotype template. Within the present first-pass summaries, the clearest localized recurrence visible in the current closed 3B/4B evidence set appears in the archive family under high tension.

This section is descriptive rather than inferential: the reported concentration is a fixed-band first-pass pattern within the archived six-packet high-tension slices, not a statistical universality claim.

5.1 A Narrow Cross-Model Concentration At Archive High Tension

Among the three closed full-family-set lines, archive is the family whose high-tension band shows the clearest narrow repeated concentration in the same break-bearing label family.

Table 6: Archive high-tension first-pass localized concentration across the closed 3B/4B lines. Counts are over the fixed six-packet high-tension band.

Model	Archive high-tension first-pass
Qwen/Qwen2.5-3B-Instruct	break-candidate (conflict-adopted)=4; surface-noise-only=2
meta-llama/Llama-3.2-3B-Instruct	break-candidate (conflict-adopted)=4; surface-noise-only=2
Qwen/Qwen3-4B	break-candidate (conflict-adopted)=6

Within the present first-pass summaries, this is the clearest localized cross-model phenotype concentration visible in the closed 3B/4B evidence set. By comparison, transcript does not show a corresponding concentration, and briefing remains mixed across the three lines.

At the same time, the concentration is narrow rather than total. The shared structure appears in the archive **high-tension** band only. The archive flat band remains model-specific:

- Qwen/Qwen2.5-3B-Instruct: compensatory-closure + conflict-respected + surface-noise
- meta-llama/Llama-3.2-3B-Instruct: surface-noise-only
- Qwen/Qwen3-4B: surface-noise + support-fused

So the result is not “archive phenotype is universal.” The narrower result is: within the present first-pass summaries and the current closed 3B/4B evidence set, archive high tension is the clearest location where the dense-transformer lines repeatedly tilt toward break-candidate structure.

5.2 Why Archive Is Different In The Present Evidence Set

This archive-specific concentration matters because it is the clearest place in the current phenotype layer where cross-model agreement becomes visible at all. Transcript and briefing remain clearly family-conditioned and model-specific. Archive, by contrast, repeatedly concentrates its high-tension packet on explicit break-bearing behavior.

This suggests that the archive family is not merely another prompt/rendering surface. On the current first-pass layer, archive is the family in which high-tension packets most often receive overt break-candidate attachment.

The report does not claim to explain why this happens mechanistically. It claims only that, within the present first-pass summaries and inside the current dense-transformer evidence set, archive high tension is the clearest recurrent phenotype concentration point.

5.3 What This Section Does And Does Not Establish

This section establishes a narrow empirical statement:

- archive high tension is the clearest cross-model phenotype concentration point visible in the current closed 3B/4B dense-transformer evidence set

It does not establish any broader claim, including:

- that archive fully stabilizes phenotype across models
- that transcript or briefing share the same localized concentration
- that break-candidate behavior is universal across all scales or architectures
- that the current archive effect explains the causal origin of hallucination or contradiction

In other words, the phenotype layer remains mostly non-portable. Archive high tension is important precisely because it is the clearest localized concentration inside that otherwise non-portable layer.

6 Protocol-Boundary Results

The main dense-transformer claim is defined not only by what it includes, but also by what it excludes. At the frozen-protocol boundary, `google/gemma-4-E2B-it` was not admitted because `transformers==4.57.3` did not recognize the `gemma4` model type, and `RWKV/v6-Finch-1B6-HF` required protocol-expanding conditions outside the frozen environment, including `trust_remote_code` and `bitsandbytes`. These are recorded as admission failures rather than as negative empirical results.

Separately, the non-transformer sidecar case was `state-spaces/mamba-2.8b-hf`. It was run only under the explicit fallback surface `prefix_mean_hidden_v1`. This was necessary because the mainline attention-weighted `Splus` extractor is not available on the same terms. Under that altered surface, the Mamba transcript line reached Gate12A replay but did not preserve the transcript-path pattern ordinarily observed on the current dense-transformer mainline. Under the altered fallback surface, the sidecar remained outside the dense-transformer mainline not only by admission rule but also by observed replay pattern: transcript replay closure did not by itself recover the same surface-specific recurrence seen on the current dense-transformer transcript path. The paper does not claim to explain why this occurs mechanistically; it records the sidecar as a boundary divergence under an explicitly changed observation surface. This is therefore treated as sidecar divergence rather than as a mainline extension.

Taken together, these boundary results delimit the main claim rather than extend it. The strongest positive evidence in the present study remains the repeated structural replay observed in the closed 3B/4B dense-transformer lines under one frozen regime. The boundary cases matter because they show where that regime fails admission or fails to preserve the same transcript-path behavior under an altered observation surface.

7 Limitations And Future Work

The present report is intentionally narrow. Its purpose is not to settle the general geometry of language-model reasoning, but to record what survives under one frozen dense-transformer regime and to mark the boundaries that become visible when that regime is stressed.

7.1 Limits Of The Current Evidence Set

The strongest positive evidence in the present study comes from closed full-family-set replay on three dense-transformer lines:

- Qwen/Qwen2.5-3B-Instruct
- meta-llama/Llama-3.2-3B-Instruct
- Qwen/Qwen3-4B

Additional dense-transformer lines broaden the structural range only at the transcript level. Those transcript-level extensions are useful for structural breadth, but they do not provide the same phenotype coverage as a closed full-family-set line.

Accordingly, the present report does not establish:

- phenotype portability across all dense-transformer scales
- full-family-set closure beyond the currently closed 3B/4B lines
- any statement about MoE architectures or frontier-scale models

7.2 Frozen FP32 As Both Strength And Ceiling

All mainline evidence is obtained under a frozen FP32 regime. This is a strength because it prevents silent environment drift, quantization-specific behavior, and architecture-specific evaluator relaxation from entering the evidence set. In the archived mainline candidate-execution manifests, this regime spans CPU execution in eight runs and CUDA execution on L4 in one run. The frozen regime is defined at the level of precision (FP32), observation surface, artifact schema, and replay rules; backend device homogeneity (CPU vs CUDA) was not imposed as an additional admission criterion.

The present report therefore does not claim a backend-equivalence theorem between CPU and CUDA execution. It claims only that, within the archived artifact path recovered here, the replayable `candidate_execution` bundle remains the scientific unit and the reported replay summaries are interpreted at that level.

At the same time, it is also a hard ceiling. The current report does not test whether the same quartet-positive structural replay survives under:

- larger dense-transformer checkpoints beyond the present frozen-FP32 local execution envelope
- multi-GPU sharded execution
- alternative precision regimes
- environment changes required by newer loader stacks

Future work should therefore treat scale extension as a controlled protocol-expansion problem rather than as a simple continuation of the current evidence line.

7.3 Architecture Coverage Remains Deliberately Incomplete

The present mainline is limited to dense transformers that preserve the frozen protocol. Non-transformer architectures are addressed only through exclusions and sidecars.

This means that the report does not establish a universal architectural law. At present, it establishes only that:

- quartet-positive structural replay persists across multiple dense-transformer lines under one frozen regime
- some candidate extensions fail admission at the protocol boundary
- a non-transformer sidecar can be run under an altered surface without preserving the current dense-transformer transcript path signature

Future work should extend this boundary mapping in two distinct ways:

- protocol-preserving extensions, where a new model is admitted without changing the frozen regime
- protocol-expanding studies, where the observation surface or loader stack is intentionally changed and the resulting evidence is kept separate from the mainline

7.4 Mechanistic Explanation Remains Open

The present study is an auditing report, not a mechanistic derivation. It demonstrates repeatable structural replay and repeatable phenotype non-portability under a declared surface, but it does not explain the internal causal mechanism that generates those outcomes.

In particular, the report does not claim:

- that loop-consistency residuals identify the causal origin of hallucination
- that trusted-tree / residual-chord decomposition is a sufficient mechanistic account of reasoning failure
- that archive high-tension localized concentration reveals a complete theory of contradiction handling

Future work should therefore connect the present macroscopic surface back to local mechanistic evidence, rather than treating the two approaches as mutually exclusive.

7.5 Immediate Next Steps

The most direct next steps are empirical rather than rhetorical.

First, the current dense-transformer evidence set should be widened by adding new full-family-set closures beyond the present 3B/4B core whenever those models can be admitted without relaxing the frozen regime.

Second, sidecar work should continue, but with explicit labeling and strict separation from the mainline. In particular, non-transformer probes are valuable precisely when they sharpen the regime boundary rather than blur it.

Future work should test whether the present first-pass phenotype layer remains stable under blinded multi-annotator relabeling and only then explore partial automation of that descriptive layer.

7.6 Final Scope Statement

The narrow claim of the present study is therefore the following:

Under a frozen FP32 dense-transformer regime, quartet-positive structural replay recurs across the closed 3B/4B full-family-set lines, while phenotype remains family-conditioned and model-specific, with the clearest localized concentration appearing only at archive high tension.

Everything beyond that statement remains open. That openness is not a weakness of the report. It is the condition that keeps the current claim controlled, stable, and falsifiable.

A Benchmark Construction

The benchmark used in this report is a fixed semi-closed retrieval-conflict benchmark over finite closed-world relational structures. The source artifact is `gate8_semiclosed_conflict_batch_v1`.

No external public benchmark such as MMLU, GSM8K, or HumanEval is used in the reported comparisons. At source level the artifact contains 200 rows. For exposition, a source item may be written as $x = (w, c, r, f, t, y)$, where w is the finite closed-world world state, c is the benchmark cell, r is the closed-world relation type, f is the retrieval rendering presented to the model, t is the answer target, and y is the defect labeling. This tuple is used here only as a compact explanatory abstraction aligned with the artifact structure; it is not intended as a literal field-by-field on-disk schema.

The 200 source rows are balanced across four explicit cells: `clean_support=50`, `direct_contradiction=50`, `distributed_incompatibility=50`, and `surface_noisy_clean=50`. The world-type counts are `genealogy=68`, `temporal=66`, and `reachability=66`. Each row keeps four layers explicit throughout construction: world truth, retrieval rendering, answer target, and defect labeling. The cell-level target regimes are fixed: `clean_support` maps to `consistent_answer` only; `direct_contradiction` maps to `consistent_answer` and `conflict_following_wrong_answer`; `distributed_incompatibility` maps to `consistent_answer` and `unsupported_bridge_answer`; and `surface_noisy_clean` maps to `consistent_answer` only.

The three families used in the report—`transcript`, `briefing`, and `archive`—are distinct rendering families over this same source benchmark. They are not separate tasks and should not be read as separate benchmark inventories. The concrete examples in Appendix A.1 are transcript-family illustrative examples from the benchmark artifacts; `briefing` and `archive` are different renderings over the same source rows and need not use identical wording.

A.1 Example Benchmark Items

The examples below are transcript-family illustrative examples from the benchmark artifacts. They are included to make the benchmark concrete for an external reader. `briefing` and `archive` are alternative renderings over the same source rows and do not necessarily preserve identical wording.

- `clean_support / genealogy / consistent_answer`. Retrieved notes: Haven is the parent of Opalx. Opalx is the parent of Umber. Warranted conclusion: Haven is an ancestor of Umber.
- `direct_contradiction / reachability / consistent_answer`. Retrieved notes: there is a directed edge from Beryl to Ivory; a cross-exam aside states that a directed path exists from Opalx to Beryl; there is a directed edge from Ivory to Opalx. Warranted conclusion: a directed path exists from Beryl to Opalx, and the cross-exam aside is not adopted.
- `distributed_incompatibility / genealogy / consistent_answer`. Retrieved notes: Yarrow is the parent of Frost. Frost is the parent of Lumen. A judge’s instruction states that separate witness fragments must not be transitively fused and that no direct ancestor relation between Yarrow and Lumen is warranted across separate ledgers. Warranted conclusion: no direct ancestor relation between Yarrow and Lumen is warranted across separate ledgers.

A fourth cell is `surface_noisy_clean`. It preserves the same warranted relation under non-substantive surface drift in the retrieved notes.

B Frozen Protocol Constants and Structural Replay Rule

The structural replay verdict is defined exactly as follows. All thresholds were fixed before any cross-model comparison and were never tuned per model or per family.

Table 7: Exact numerical constants and structural-replay decision rule.

Constant / Rule	Exact value	Operational meaning
$\tau_{\text{overlap_sv_min}}$	1.0×10^{-8}	Overlap singular values must exceed this threshold to count as nonzero active rank.
$\tau_{\text{transport_gap_fro}}$	1.0×10^{-6}	An edge is transport-admissible only if its transport-gap Frobenius residual $\leq \tau$ and the overlap-rank check passes.
$\tau_{\text{holonomy_residual_fro}}$	1.0×10^{-6}	A triangle cycle is compatible only if its composed holonomy residual $\leq \tau$.
balanced-per-band	6	High-tension band size used in archive localized-concentration counts (Table 6).
Structural quartet	$q_1 \wedge q_2 \wedge q_3 \wedge q_4$	q_1 : zero-overlap clearance; q_2 : all defined triangles anchor-rich; q_3 : trusted-tree median $>$ residual-chord median; q_4 : plain median $>$ anchor-qualified median.
Structural replay verdict	quartet positive	A run is counted as structural replay only when all four q_i are jointly true under the frozen protocol.

C Gate8 Candidate-Execution Conditions (Frozen Mainline)

The archived mainline candidate-execution artifacts, including the per-sample `meta.json` records, recover the following exact conditions. Stochastic decoding parameters were deliberately not used because the path is teacher-forced.

Table 8: Exact Gate8 execution conditions recovered for the frozen mainline path.

Field	Fixed value	Note
<code>seed</code>	7	Fixed across all archived runs
<code>topk_requested</code>	128	Extractor top- k size
<code>extraction_mode</code>	<code>teacher_forcing_forward_v1</code>	Teacher-forced over fixed answer targets
<code>deterministic_requested</code>	<code>true</code>	Deterministic extractor mode; CUDA-specific controls apply when CUDA is used
<code>Splus</code>	<code>attn_lastlayer_weighted_hidden_v1</code>	Frozen mainline observation surface
<code>Sminus</code>	<code>lm_head_row_expectation_topk128_v1</code>	Negative-support definition
<code>temperature / top-p / do_sample</code>	N/A	Not applicable (teacher-forced path)

Across the archived closed 3B/4B mainline candidate-execution manifests, eight runs record CPU execution and one records CUDA execution on L4. The scientific unit remains the replayable `candidate_execution` artifact regardless of execution device.

D Artifact Manifest

The Zenodo package is organized into three artifact tiers.

- closed mainline replay artifacts for the three 3B/4B dense-transformer lines
- supporting transcript-level extension artifacts, including the imported Mistral-7B transcript bundle
- boundary artifacts documenting frozen-protocol exclusions and the Mamba sidecar divergence

Within the package, each archived run preserves its own manifest, checksum, and per-sample meta records. The main text reports only the aggregate summaries needed for the present claim; readers seeking exact run-level provenance should consult the corresponding manifest, checksum, and per-sample meta entries within each archived artifact directory.

E Supporting Transcript-Level Structural Observations

Table 9 summarizes the dense-transformer lines that support the transcript-level structural extension discussed in Section 4.2, from 360M upward and including one imported 7B transcript bundle. These runs are included as supporting observations rather than as mainline closed-family-set phenotype evidence.

Table 9: Supporting transcript-level structural observations outside the mainline closed 3B/4B family sets.

Model	Source	Evidence status	Structural quartet
SmolLM2-360M	local	transcript only	positive
meta-llama/Llama-3.2-1B-Instruct	local	transcript only	positive
Qwen/Qwen2.5-1.5B-Instruct	local	transcript only	positive
mistralai/Mistral-7B-Instruct-v0.3	imported	transcript only	positive

F Reproducibility Note

The Zenodo release archives both the fixed-protocol implementation and the mainline artifacts used in the present comparison.

- mainline artifact-generation commit recorded in the closed 3B/4B manifests: 084eb7878d8cb016243950e1cf4b4bd7379daaba
- archived frozen reference implementation and sidecar-boundary commit: 8a14c94b999d823c1734b2582730c3ad4ea98d03

- archived mainline artifact roots:
 - `runs/gate12a_cross_model_replay_qwen.qwen2.5_3b_instruct/`
 - `runs/gate12a_cross_model_replay_meta.llama.llama.3.2_3b_instruct/`
 - `runs/gate12a_cross_model_replay_qwen.qwen3_4b/`
- each archived run preserves its own manifest, checksum, and per-sample meta records; these manifests and per-sample meta records are the source of truth for the exact Gate8 execution fields and bundle-level run settings summarized in Section 2 and Appendix C
- mainline hardware/software setting:
FP32 execution across the archived mainline candidate-execution runs (CPU for 8/9 runs, CUDA on L4 for 1/9 run); `transformers==4.57.3`
- The downstream Gate12A cross-model replay summaries and structural quartet verdicts are frozen under commit `084eb7878d8cb016243950e1cf4b4bd7379daaba`. One upstream Gate8 candidate-execution artifact (the `Qwen/Qwen2.5-3B-Instruct` transcript bundle) was generated under an earlier commit `58d06742f23a0bc7ba25c6ecde790e2e03b4324e`; all other closed mainline Gate8 bundles align with the mainline commit.
- imported Mistral provenance: `mistralai/Mistral-7B-Instruct-v0.3` appears only as an imported local-GPU `transcript_v1` bundle under `runs/remote_gpu_imports/mistral7b_transcript.2026-04-03/`, with replay summary under `core/runs/gate12a_cross_model_replay_mistral_7b_instruct_v0_3/`; it is supporting transcript-level evidence only, not a locally generated full-family-set mainline run and not full-family phenotype evidence.

References

- [1] Catherine Olsson, Nelson Elhage, Neel Nanda, Nicholas Joseph, Nova DasSarma, Tom Henighan, Ben Mann, Amanda Askell, Yuntao Bai, Anna Chen, et al. In-context learning and induction heads. *arXiv preprint arXiv:2209.11895*, 2022.
- [2] Arthur Conmy, Augustine N. Mavor-Parker, Aengus Lynch, Stefan Heimersheim, and Adrià Garriga-Alonso. Towards automated circuit discovery for mechanistic interpretability. *arXiv preprint arXiv:2304.14997*, 2023.
- [3] Michael M. Bronstein, Joan Bruna, Taco Cohen, and Petar Veličković. Geometric deep learning: Grids, groups, graphs, geodesics, and gauges. *arXiv preprint arXiv:2104.13478*, 2021.
- [4] Tianyi Zhao, Yinhan He, Wendy Zheng, Yujie Zhang, and Chen Chen. Wired for overconfidence: A mechanistic perspective on inflated verbalized confidence in LLMs. *arXiv preprint arXiv:2604.01457*, 2026.